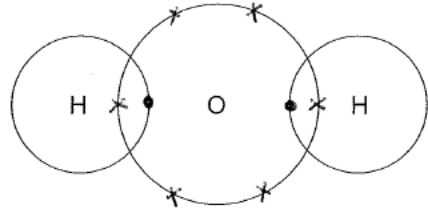


Question				Marking details	Marks available												
					AO1	AO2	AO3	Total	Maths	Prac							
4	(a)	(i)		 award (1) for two shared pairs of electrons award (1) for complete octet in oxygen		2		2									
		(ii)	<table><tr><td>poor conductor of electricity</td><td></td></tr><tr><td>colourless</td><td></td></tr><tr><td>good conductor of heat</td><td></td></tr><tr><td>low melting point and boiling point</td><td>✓</td></tr></table>	poor conductor of electricity		colourless		good conductor of heat		low melting point and boiling point	✓	1			1		
poor conductor of electricity																	
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Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)	I	all points plotted correctly (1) tolerance $\pm \frac{1}{2}$ small square straight line passing through all points (1) do not penalise missing label		2		2	2	
			II	straight line passing through (0,0) and (10,25) do not penalise missing label			1	1	1	
			III	volume of hydrogen is double / twice the volume of oxygen (2) award (1) for either of following - as the volume of oxygen increases the volume of hydrogen increases - reading from graph given e.g. 50cm³ of hydrogen formed with 25cm³ of oxygen			2	2	2	2
		(ii)		D			1	1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		plastics are (electrical) insulators / do not conduct (electricity)	1			1		2
		(ii)		it is negatively charged / it is the cathode (1) accept negative opposite charges attract / positive ions are attracted to negative electrode (1) neutral answer - it is attracted	2			2		2
		(iii)		$2\text{Ag}^+ + 2\text{e}^- \longrightarrow \text{Ag}$ ☐ $\text{Ag}^+ + \text{e}^- \longrightarrow \text{Ag}$ ☒ $\text{Ag}^+ - \text{e}^- \longrightarrow \text{Ag}$ ☐ $\text{Ag}^+ + 2\text{e}^- \longrightarrow \text{Ag}$ ☐		1		1		
				Question 4 total	4	5	4	13	5	6